SESSION 3 LECTURE NOTES

Integrating Network KPIs with Geospatial Data: Context-Aware Telecom Infrastructure

GeoAI Fusion Bootcamp 2026

Introduction

Traditional network analysis treats performance metrics as numbers in a spreadsheet, disconnected from the physical and human environment surrounding each cell site. Key performance indicators, such as throughput, latency, and call-drop rate, describe symptoms without explaining their cause. Engineers reading a table of numbers must react to problems without understanding why a given site underperforms.

The problem this session addresses is one of blindness rather than data scarcity. Telecom operators already collect enormous volumes of network telemetry, but a spreadsheet cannot reveal a geographic blind spot. Plotting the same throughput, latency, and call-drop figures against geographic location instantly shows where performance degrades, and it allows engineers to correlate poor key performance indicators with terrain or interference in ways a raw table never reveals.

This session's objective is to demonstrate how integrating network key performance indicators with multi-sensor geospatial data, described in this bootcamp as GeoAI, transforms telecom operations from reactive troubleshooting into predictive, intelligent planning. Participants examine three layers of contextual data, namely network coverage, physical terrain, and human demand, before considering how these layers combine into a unified decision framework.

Discussion

From Raw Tables to Spatial Revelation

A spreadsheet listing site coordinates, throughput, and drop rate forces an engineer to infer geography mentally from raw numbers. As an illustration, a table might show several sites clustered in one area with throughput above one hundred megabits per second and drop rates below one per cent, while a separate cluster of sites nearby records throughput below thirty megabits per second and drop rates above five per cent. Plotting these same values on a map instantly shows that the second cluster falls within a specific zone, prioritising which sites need investigation first, an insight the table alone could never surface efficiently.

Layer One: Mapping the Network's True Coverage and Trends

Coverage visualisation exposes marginal-coverage pockets that customer complaints alone would miss. Correlating a dead zone with building data explains whether the cause is a line-of-sight obstruction or an equipment fault, allowing engineers to target the correct root cause rather than guessing. Spatially aware trend analysis tracks key performance indicator shifts across zones and seasons, revealing patterns such as downtown congestion concentrated during commuting hours or coastal signal degradation during the

monsoon season. Longitudinal analysis of this kind allows operators to plan capacity upgrades ahead of demand rather than after a failure occurs.

Layer Two: The Physical Landscape and Signal Propagation

Terrain and land cover attenuate radio signals differently across frequency bands. Hills, dense vegetation, and urban geometry all block or weaken a signal in ways that vary by location. Digital elevation models, drawn from Earth observation sources such as the Shuttle Radar Topography Mission or the Copernicus DEM, allow planners to simulate real-world coverage before a tower exists. Simulating signal propagation on paper catches coverage problems before construction begins, avoiding an expensive post-deployment failure.

Transient Weather and Invisible Network Dips

Heavy rainfall, humidity, and storms degrade signal quality noticeably, particularly at the higher frequency bands used by modern technologies such as fifth-generation networks. Correlating historical climate products, such as ERA5 or CHIRPS precipitation data, with network performance logs allows an operator to distinguish a genuine hardware fault from a transient, weather-driven dip. This distinction changes how urgently a repair crew is dispatched to a reported outage, since a weather-driven dip typically resolves itself once the storm passes.

Layer Three: Aligning Infrastructure with Human Reality

Capacity planning must follow where people actually gather, not simply where towers currently stand. Fusing high-resolution population grids and mobility data with network key performance indicators identifies high-density areas approaching capacity limits before users experience congestion. Understanding local income levels and typical data-consumption patterns further helps an operator forecast realistic demand and price services appropriately, avoiding a rollout strategy that assumes every location behaves identically.

Identifying Underserved Gaps for Targeted Investment

Overlaying coverage maps, favourable terrain, and socio-economic indicators surfaces communities that are both poorly served and possess genuine unmet demand. This intersection of network coverage, physical terrain, and socio-economic context represents the zone where targeted capital investment delivers the highest commercial and social return. Regulatory bodies increasingly rely on this same socio-economic fusion to guide universal-service funding and policy prioritisation.

Harmonising the Data Stack

Geospatial fusion in telecom fails if the underlying layers do not align, exactly as described in Session 2. Sentinel-2 imagery at ten metres, Sentinel-1 imagery at twenty metres, digital elevation data at thirty metres, and population data at yet another resolution must all be resampled, reprojected, and aligned before they can combine. The result is a unified, multi-layer data cube sharing the same projection, resolution, and time frame. Consistent scales, unified coordinate reference systems, and precise cloud masking together create the foundation required for a reliable GeoAI model in the telecom domain.

From Reactive to Predictive Operations

A comparative diagnostic clarifies the shift this session describes. Traditional troubleshooting reacts to customer complaints and arbitrary key-performance-indicator drops. GeoAI-fused operations instead pinpoint physical causes instantly by correlating multiple data layers. Traditional capacity planning forecasts demand solely from past utilisation at a given tower. GeoAI-fused planning instead forecasts demand from human mobility and income data. Traditional maintenance waits until infrastructure is already damaged. GeoAI-fused maintenance instead uses predictive response, drawing on weather data to pre-position repair crews ahead of an expected failure.

The Unified Geospatial Telecom Framework

True network intelligence does not emerge from any single dataset. It emerges exclusively from the intersection of network key performance indicators, Earth observation data, and socio-economic data, feeding into a decision engine that produces predictive analytics, optimised routes, and real-time network adjustments. The outcomes that follow include resilient infrastructure, reduced environmental impact, and stronger long-term planning. This framework bridges the gap between the technical metrics reported by the machine and the physical reality experienced by the user.

Real-World Applications

Flood-aware site planning illustrates the value of this framework directly. Siting critical infrastructure in an unmapped, seasonal flood zone guarantees future outages and a substantial repair bill. Overlaying historical flood-extent data, derived from Earth observation imagery, onto a proposed site allows a planner to adjust the location before concrete is poured. Preventive spatial analysis of this kind is far cheaper than relocating or repairing a water-damaged site after the fact. Predictive outage response offers a second illustration. Fusing real-time weather feeds with historic network performance data enables an operator to anticipate exact outage locations and pre-position repair crews ahead of extreme weather, rather than waiting for a customer complaint. This measurably reduces both network downtime and the volume of customer-reported complaints, protecting brand reputation in the process.

Conclusion

This session demonstrated that a telecom network cannot be understood or optimised in isolation from its physical and human context. Integrating network key performance indicators with terrain, climate, population, and income data converts a reactive troubleshooting process into a predictive planning discipline capable of anticipating both outages and unmet demand.

The future of telecommunications infrastructure in ASEAN and comparable regions requires treating the network not as a closed circuit, but as a living system embedded in the physical and human world. Mastering the geospatial data engineering and feature-extraction techniques introduced in this session is the critical step toward building infrastructure that is resilient, profitable, and equitable.